

MIDI Maze Gameplay Communication Protocol

This document describes the on-wire communication used by MIDI Maze during live gameplay. It covers the transition from setup into gameplay, the steady-state gameplay communication loop, controller and payload formats, resynchronization behavior, and error handling. Gameplay simulation and rendering are outside the scope of this document.

1. Transition from Setup to Gameplay

After all machines complete setup and configuration exchange, the Master issues CMD_START_GAME (\$87). Upon receipt, all machines exit setup mode and enter gameplay initialization.

Before live gameplay begins, MIDI Maze performs a structured ring-wide synchronization exchange to ensure that all machines share identical gameplay parameters.

Gameplay Entry Relay (7-Byte Parameter Block)

Byte	Semantic Name	Description
0	FireCooldown	Weapon fire cooldown duration used by the firing state machine.
1	ReloadTimer	Weapon reload or phase timing parameter.
2	ProjectileLife	Projectile lifetime or timing parameter.
3	WeaponMode	Weapon mode selector used by firing logic.
4	MoveSpeedFlag	Movement speed modifier flag.
5	ProjectileSpeedFlag	Projectile speed modifier flag.
6	TurnRate	Turn rate step size for player rotation.

2. Steady-State Gameplay Communication Loop

Once gameplay begins, communication becomes a continuous byte stream relayed around the ring. Each machine injects its own input updates while forwarding all received bytes. There is no packet framing; byte order and position define meaning.

Normal Update: One controller byte

Extended Update: Controller byte followed immediately by a payload byte

3. Controller Byte Format

Bit	Name	Meaning
0	JoyUp	Joystick Up pressed
1	JoyDown	Joystick Down pressed
2	JoyLeft	Joystick Left pressed
3	JoyRight	Joystick Right pressed
4	Fire	Trigger pressed
5	Reserved	Unused
6	Reserved	Unused
7	ExtendedFlag	Indicates that a payload byte immediately follows this controller byte.

4. Extended Payload Byte

The payload byte carries non-movement data such as text input or command codes. It is only valid when preceded by a controller byte with the ExtendedFlag set.

Value Range	Type	Meaning
\$00–\$7F	Text / Control	Character input or control actions such as Return or Backspace.
\$80–\$FE	Command	In-game command dispatched by gameplay logic.
\$FF	Null	No payload or default state when no text or command is active.

5. Confirmed In-Game Payload Commands

Command	Name	Description	Phase
\$81	ClearState	Clears shared per-player state arrays used for transient gameplay data.	Gameplay
\$82	HoldSync	Displays 'PLEASE HOLD' and enters a synchronization wait state.	Setup & Gameplay
\$84	Resync	Initiates a ring-wide resynchronization exchange. Any machine may request resync; the Master performs the broadcast while slaves receive and apply state.	Setup & Gameplay
\$86	RosterExchange	Initiates a structured ring exchange of player-related data such as names or roster information.	Gameplay

6. Resynchronization (\$84) Process

Resynchronization is a modal protocol phase used to realign gameplay state. Any machine may request resync by transmitting \$84 as an extended payload command.

When the Master receives \$84, it drives the resynchronization process by transmitting a marker byte (\$83) followed by a fixed configuration payload. All other machines enter a receive-only role and wait for the marker before accepting data.

Resynchronization depends on timely ring circulation. If communication delays exceed internal deadlines, the resync attempt fails and gameplay terminates.

7. Error and Status Codes (L3ED2)

Code	Message	Phase
\$02	Game terminated	Setup & Gameplay
\$03	Maze too small!	Setup
\$04	Network error	Setup
\$05	Too many machines!	Setup
\$06	No drones allowed!	Setup
\$07	Checksum error	Setup
\$08	Device not responding	Setup
\$09	No such device	Setup
\$0A	Can't sync	Setup
\$C7	I/O timeout	Gameplay

8. Game Termination Behavior

MIDI Maze does not transmit an explicit end-of-game command. Gameplay ends due to error, timeout, or command-driven transition. The game prints a status message and returns to the main menu.